Jinhao Li Ph.D.

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Education

Shanghai Jiao Tong University, China 2021–2027.03 *(Estimated)*
Ph.D. in Mechanical Engineering Advisor: Prof. Guoying Gu

Hunan University, China 2017–2021
B.Eng. in Vehicle Engineering Advisor: Prof. Huigao Duan

Research Interest

- Iontronic meta-interfaces and biomedical sensory systems
- Soft miniature machines for adaptive tissue interaction

Research Experience

Ph.D. Thesis Topic: Three-dimensional iontronic sensors at human-robot interfaces

Iontronic sensors facilitate high-sensitivity conformal detection, while their capacity to resolve complex deformations remains constrained by thin-film architectures. Herein, I proposed a multimaterial cryogenic printing technique for sophisticated hydrogel structures by harnessing an all-in-cryogenic solvent phase transition strategy. I further designed a series of 3D-architected hydrogel meta-structures and decoupling mechanisms to program iontronic sensing behaviors in high-fidelity proprioception and tactile sensing. As prototypes, I developed self-sensing artificial heart valves, somatosensory soft neuroprosthetic hand, and wireless infection-warning bandages, enriching smart bionics at human-robot interfaces.

B.Eng. Thesis Topic: Multiphysics-effected iontronic behaviors in flexible batteries

Flexible batteries facilitate epidermal self-powered healthcare but suffer from dendrite growth and thermal runaway. Herein, I developed an ultrathin zinc lattice electrode with biomimetic microhole arrays, enabling spatially selective deposition at engineered iontronic interfaces. I further proposed a zone-based design optimization approach for battery thermal management to suppress performance degradation. My multiphysics regulation strategy delivers 2,000-hour cycle stability in flexible cells, paving the way for wearable healthcare.

Research Skills

Device development:
- Design: Structure modeling and optimization, Multimodal sensing, Iontronic interfaces
- Analysis: Multiphysics simulation, Kinetic modeling, In-situ characterization
- Polymer development: Conducting polymers, Hydrogels, Elastomer hybrids

Cleanroom micro-fabrication:
- Material growth: ALD, Sputtering, Thermal evaporation, Electro-/electroless-plating
- High-resolution patterning: Photolithography, Multimaterial 3D printing

System integration:
- Wearable electronics: Wireless electronic skin, Stretchable circuits, Flexible batteries
- Sensorized prosthesis: Somatosensory robot, Human-machine interaction, Sensory feedback loop

Publications and Patents

1. 3D-printed PEDOT:PSS for soft robotics.
   Jinhao Li[#], Jie Cao[#], Baoyang Lu*, Guoying Gu*.
   Nature Reviews Materials, 8, 604-622 (2023).
2. Multimaterial cryogenic printing of three-dimensional soft hydrogel machines.
   Jinhao Li, Jie Cao, Rong Bian, Rongtai Wan, Xiangyang Zhu*, Baoyang Lu*, Guoying Gu*.
   Nature Communications, 16, 185 (2025).
3. A wireless skin-mimetic bandage for continuous thermomechanical monitoring of periprosthetic joint infection.
   Jinhao Li[#], Jialian Xu[#], Ting Xie, Yanzhe Zhao, Dengfeng Li, Xinhua Qu*, Guoying Gu*.
   Proceedings of the National Academy of Sciences of the United States of America, Submitted (2026).

4. Soft 3D lattice iontronic tactile sensor.
   Jinhao Li, Qinghua Yu, Zequn Shen, Dong Wang, Guoying Gu*.
   SmartBot, 1, e70002 (2025).
5. Large-scale zone-based approach to global modeling and optimization for a novel thermal management system of module-free lithium-ion battery.
   Jinhao Li, Fei Lei*, Wenhao Zhu, Zhifeng Qiu, Hailong Wu.
   Structural and Multidisciplinary Optimization, 64 3621-3636 (2021).
6. Ultrathin and ultralight Zn micromesh-induced spatial-selection deposition for flexible high-specific-energy Zn-ion batteries.
   Huaizhi Liu[#], **Jinhao Li**[#], Xianan Zhang, Xiuxue Liu, Yu Yan, Fengjun Chen, Guanhua Zhang*, Huigao Duan*.
   Advanced Functional Materials, 31, 2106550 (2021).
7. Biomimetic honeycomb Zn anode enabled multi-field regulation toward highly stable flexible Zn-ion batteries.
   Huaizhi Liu, **Jinhao Li**, Donghai Wei, Xiuxue Liu, Zheren Cai, Hang Zhang, Zhisheng Lv, Lei Chen, Haicheng Li, Hongyu Luo, Yanli Zhao, Huihuang Yu, Xiaohu Wang, Fengjun Chen, Guanhua Zhang*, Huigao Duan*.
   Advanced Functional Materials, 33, 2300419 (2023).
8. Robust mechanically interlocked network ionogels.
   Mengling Yang, **Jinhao Li**, Chunyu Wang, Li Yang, Zhiwei Fan, Wenbin Wang, Guoquan Liu, Shaolei Qu, Zhaoming Zhang, Jiang Zou, Wei Yu, Guoying Gu*, Xuzhou Yan*.
   Angewandte Chemie International Edition, 64, e202423847 (2025).
9. High-stretchability, ultralow-hysteresis conducting polymer hydrogel strain sensors for soft machines.
   Zequn Shen, Zhilin Zhang, Ningbin Zhang, **Jinhao Li**, Peiwei Zhou, Faqi Hu, Yu Rong, Baoyang Lu*, Guoying Gu*.
   Advanced Materials, 34, 2203650 (2022).
10. Millimeter-scale fluid-driven soft robots.
    Rong Bian, Ningbin Zhang, Xinyu Yang, **Jinhao Li**, Dezhi Yang, Jieji Ren, Jiang Zou, Guoying Gu*.
    National Science Review, 12, nwaf413 (2025).
11. Three-dimensional Fabrication of hydrogels. China Patent, CN202411841590.0.
    Guoying Gu, **Jinhao Li**.
12. Fabrication of flexible patternable Zinc electrode. China Patent, CN202010532726.5.
    Guanhua Zhang, **Jinhao Li**, Huigao Duan, Jin Hu.

Grants, And Awards

Principal Investigator, Young Scientists Fund for Doctoral Students (Grant No. 524B2047)
National Natural Science Foundation of China (NSFC) 2024-2026

Youth Talent Support Program for Doctoral Students
China Association for Science and Technology (CAST) 2025-2026

National Scholarship (Undergraduate and Ph.D.)
Ministry of Education of China 2025, 2018

Graduate Academic Rising Star (10 recipients annually)
The 10th Chinese Soft Robotics Conference 2025

Best oral presentation award
IROS Workshop Tactile Sensing Toward Robot Dexterity and Intelligence 2025

Graduate Academic Rising Star Finalist (Top 1‰)
Shanghai Jiao Tong Univerisity 2025

Doctoral Graduate with Distinction
Shanghai Jiao Tong Univerisity 2026

Talks

Cryogenic printing of three-dimensional hydrogel machines
Embodied Intelligence Conference March, 2026